DSA CA 1 - REPORT / PROPOSAL

BSHC – Year 2

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Theme

The theme I've decided to choose from the Smart Docklands is **Urban Greening/Biodiversity**[1] and I've decided to focus on developing a simple pilot application that encourages people to participate in "biodiversity conservation and sustainable practices". The idea is for them to have access to this platform to create awareness and allow us to study "what drives engagement" from their interaction with the system. This pilot application shall be called "EquipWise".

Application Overview

Citizens Information[2] explains how **reusing** waste can significantly reduce waste. EquipWise gives users access to this option, allowing them to advertise their second-hand sports equipment and sell them at their chosen price, eventually increasing biodiversity. Sports equipment is a significant concern regarding wastage. To promote sustainability, EquipWise will focus on second-hand sports equipment, specifically archery equipment[3] as I have some knowledge on archery and have been searching for second-hand archery equipment for our club at NCI. Therefor this could be a way for future sports clubs to source and sell affordable second-hand equipment, extending the lifecycle of these products and reduces waste generated by sporting activities[4].

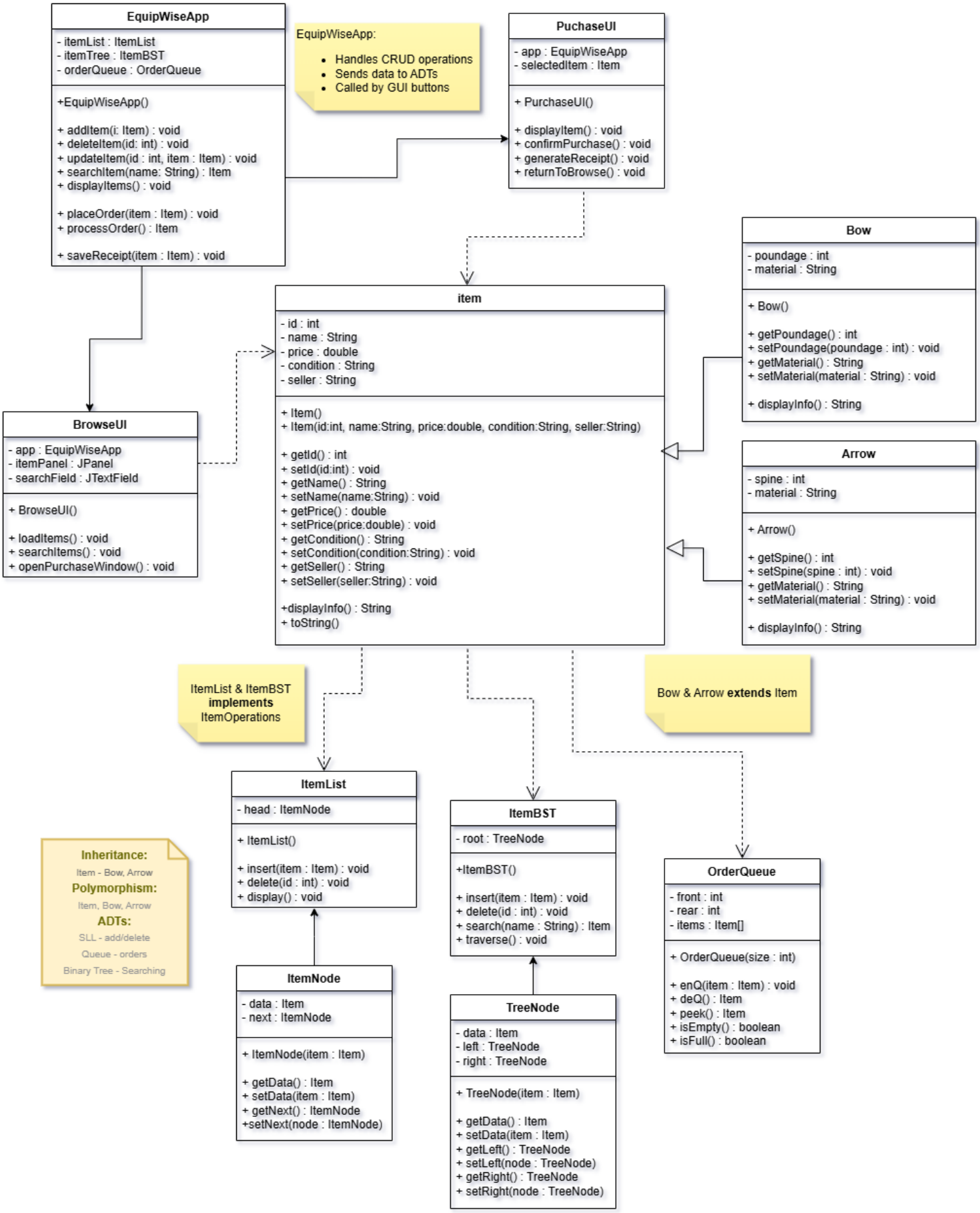
This pilot version of EquipWise will be developed in **NetBeans** using Java, with drag-and-drop Swing GUI and will feature two main interfaces that users can witch between. The first interface is a browsing window where users can view and search available equipment listings. This interface will also display product/item information such as name, condition, price and seller. The second interface allows users to confirm a purchase request for a selected product. This interface keeps the system simple and user-friendly.

From a technical perspective, the application will demonstrate the use **of abstract data types** and algorithms. Equipment items will be stored in a **Singly Linked List** which allows users to add, remove and display items. A **Binary search Tree** will allow users to efficiently search for items and a **Queue** will manage purchase requests, allowing orders to be processes in the order they were received (FIFO). Once a purchase is confirmed, a receipt will be generated and saved in a **.txt** file.

The development process of EquipWise will be tracked using a public **GitHub** repository "**EquipWisePilotApp**".

(*GitHub Repository: https://github.com/IA05/EquipWisePilotApp.git*)

EquipWise Class Diagram



References

[1] Smart Docklands, “Smart Docklands Call for Pilots 2025: Deadline Extended.” Accessed: Mar. 15, 2026. [Online]. Available: https://smartdocklands.ie/smart-docklands-call-for-pilots-2025-deadline-extended/

[2] Citizens Information, “Reducing Waste.” Accessed: Mar. 10, 2026. [Online]. Available: https://www.citizensinformation.ie/en/environment/waste-and-recycling/reducing-waste/

[3] International Bow, “Try not to break it: Sustainability in archery.” Accessed: Mar. 15, 2026. [Online]. Available: https://www.bow-international.com/features/try-not-to-break-it-sustainability-in-archery/

[4] Stanley John and Archery World, “Archery consumables and recycling: Nocks, vanes and arrow shafts.” Accessed: Mar. 10, 2026. [Online]. Available: https://www.worldarchery.sport/news/171469/archery-consumables-and-recycling-nocks-vanes-and-arrow-shafts